PROJECT DOCUMENTATION

01

User Manual

How visitors, the owner and AirShare users work with rishavprasad.in — every page, the contact form, file transfer and the private content dashboard.

PREPARED FOR	SITE	VERSION	DATE
Rishav Prasad Owner and maintainer	rishavprasad.in Cloudflare Workers	1.0	31 October 2026

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User Manual · version 1.0 · 10 sections

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01

About this manual

This manual explains how the public portfolio behaves and how its owner operates it day to day. It is written for the site owner first, and for anyone who takes over maintenance later.

The site is a static Next.js export served by Cloudflare Workers, with a small API for the contact form, content editing and AirShare. Public pages are generated at build time; the content dashboard writes to a Cloudflare D1 database, and a publish action asks GitHub to rebuild the site.

Who this manual is for

- **Visitors** — read sections 02 to 05 and 07 to learn what each page offers.
- **The owner** — read section 06 for the dashboard workflow and sections 08 to 09 for troubleshooting and data handling.
- **A future maintainer** — start here, then read the Technical Specification (02) and the File Reference (06).

Status labels used across the documentation

LABEL	MEANING
WORKING	Implemented and verified in this engagement, with evidence.
PARTIALLY WORKING	Works, but with a known constraint or an unverified path.
NOT IMPLEMENTED	Not built; no claim is made that it exists.
REQUIRES CONFIGURATION	Depends on an account setting, secret or external service.
NOT TESTED	Cannot be verified without credentials or a manual action by the owner.

What is deliberately not covered

Server-side Cloudflare account settings that the local API credentials cannot read are reported as “requires verification” rather than guessed. The admin dashboard’s signed-in screens are not shown in this package because they need the owner’s password; the sign-in screen is shown instead.

02

The portfolio at a glance

The site has a single public navigation with eight sections, a contact page, an AirShare guide and a browser app. Every page is server-rendered to static HTML at build time and hydrated with React for theme switching and interactive widgets.

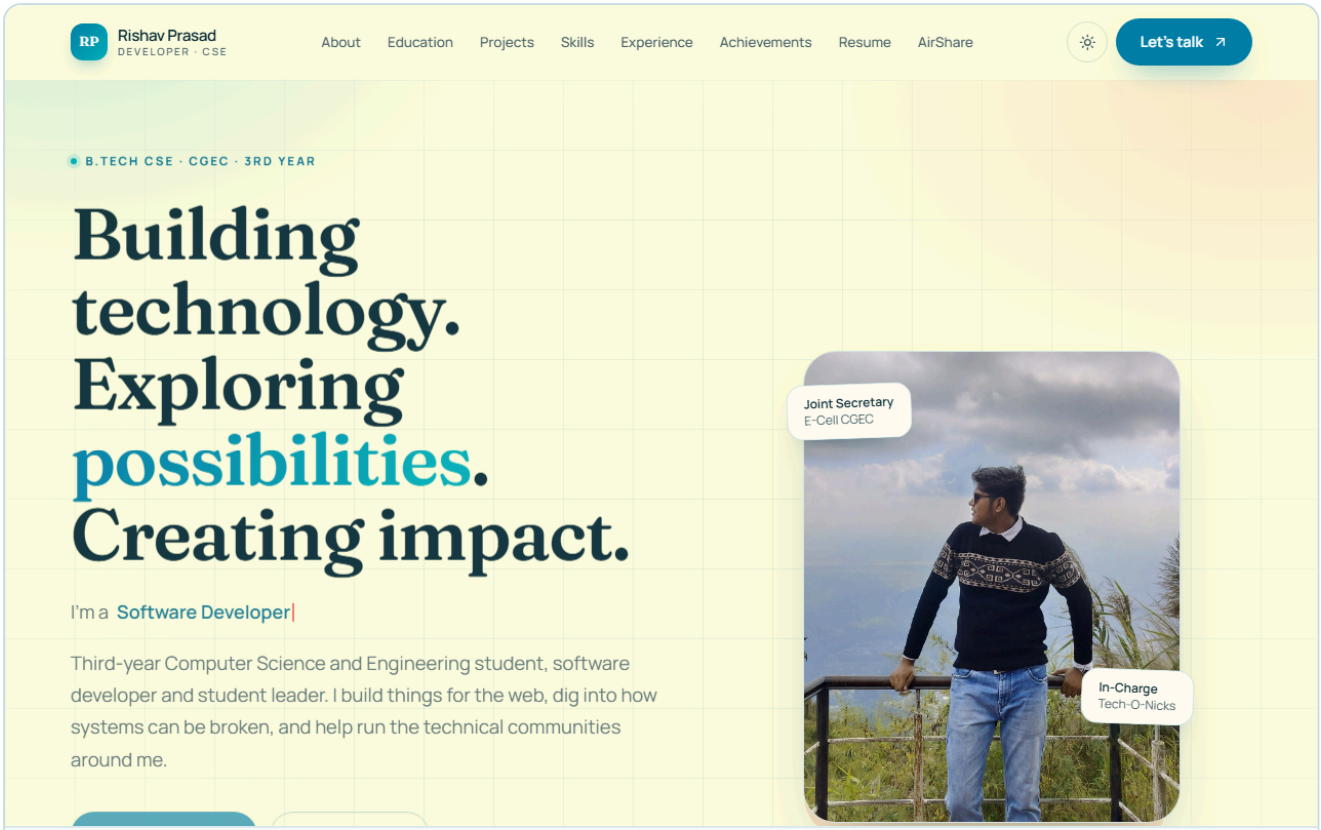


Figure 1. The home page at a desktop width (1280 × 800). The portrait, role animation and leadership cards are part of the static build.

Route map

ADDRESS	PAGE	PURPOSE
`/`	Home	Introduction, focus areas, leadership highlights and contact call to action.
`/about/`	About	Biography, journey timeline and short personal summary.
`/education/`	Education	B.Tech and school records with scores.
`/projects/`	Projects	Project cards with summaries, stacks and links.
`/skills/`	Skills	Three grouped skill areas.
`/experience/`	Experience	Leadership roles and responsibilities.
`/achievements/`	Achievements	Leadership records, awards and certifications.
`/resume/`	Resume	On-page resume summary with the approved PDF.
`/airshare/`	AirShare guide	Explains the LAN-only transfer tool and its limits.
`/airshare/app/`	AirShare app	PIN-gated browser-to-browser file and message transfer.
`/contact/`	Contact	Message form with Cloudflare Turnstile, plus email.
`/admin/`	Admin	Private content dashboard; excluded from search engines.
any unknown path	Not found	Custom 404 page with links back into the site.

The sitemap lists the eleven public URLs; `/admin/` and the 404 page are intentionally excluded.

Mobile layout

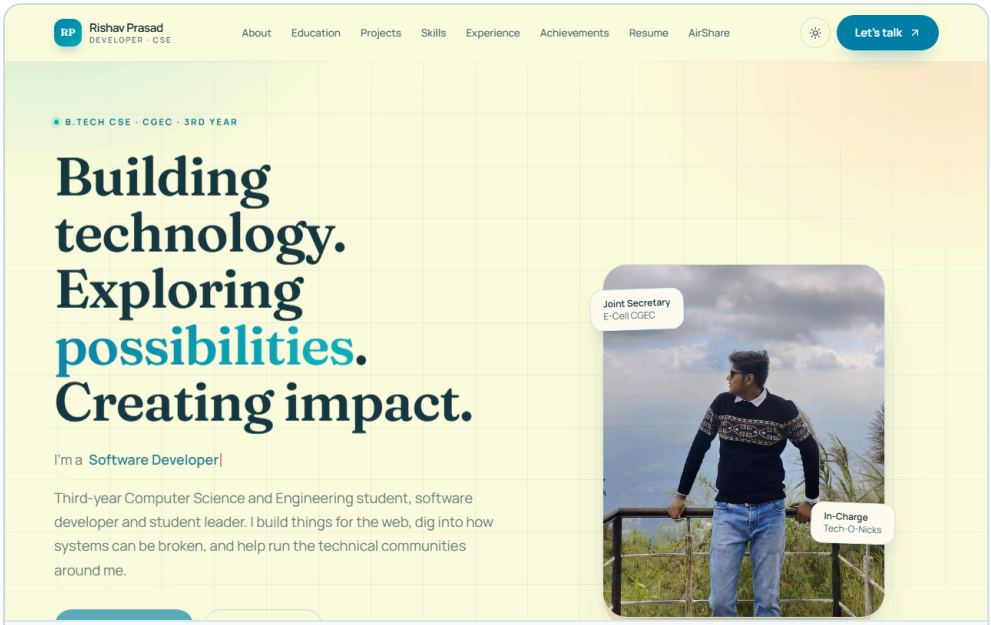


Figure 2. Desktop layout. The navigation collapses below the large breakpoint.



Figure 3. Phone layout at 375 px, with a menu button.

03

Browsing the public pages

All public pages share the same header, footer and theme control. The header links to About, Education, Projects, Skills, Experience, Achievements, Resume and AirShare, with a “Let’s talk” button to the contact page. On narrow screens the links move into a menu panel.

Home

The home page opens with the name, the current programme (B.Tech Computer Science and Engineering at CGEC), an animated role line and a portrait. It then presents four focus areas — software development, web development, cybersecurity and artificial intelligence — followed by three leadership cards: E-Cell CGEC, Tech-O-Nicks and the Smart India Hackathon 2026 organising team. The closing panel invites enquiries by email or through the contact form.

About

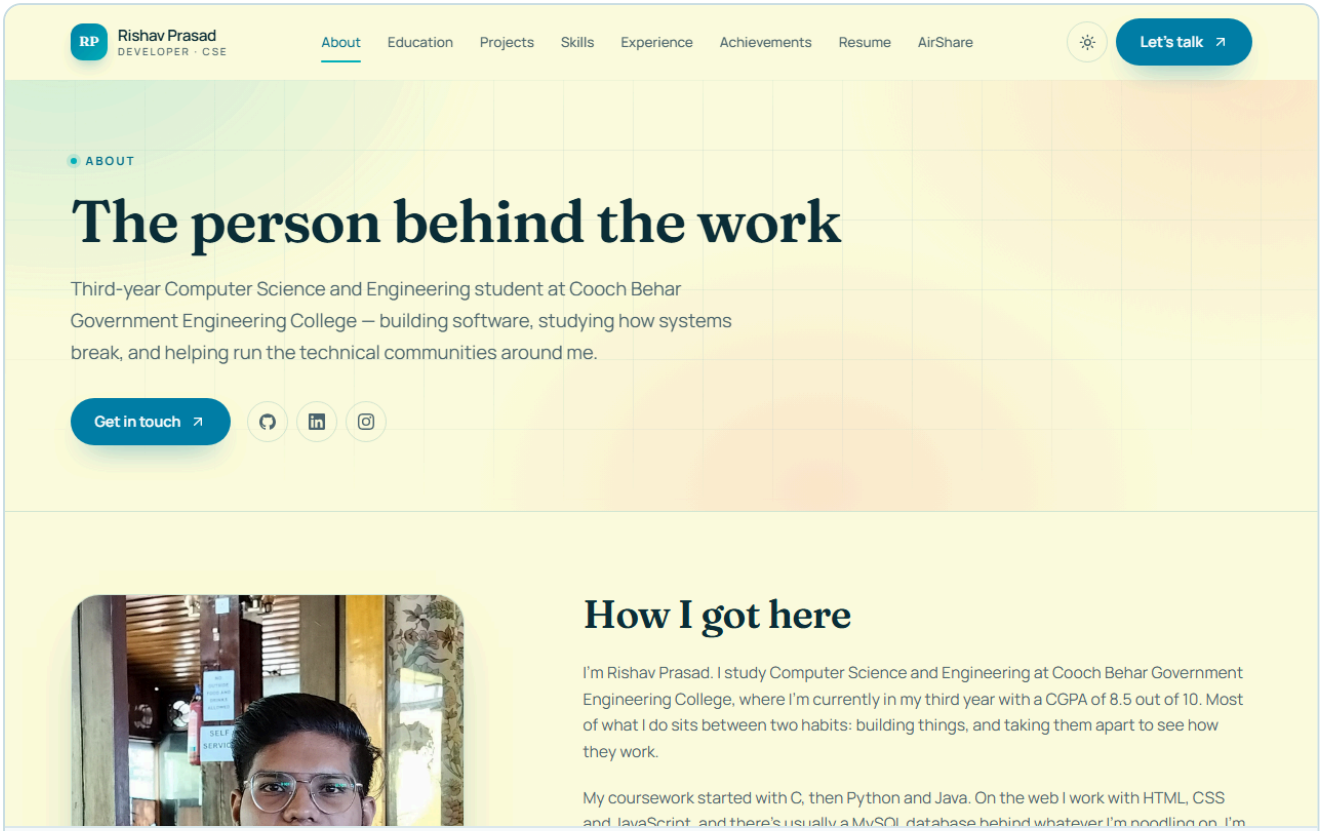


Figure 4. The About page combines the biography, a chronological journey and a "short version" summary.

The About page explains the background, how the interest in building software developed, and the principles behind the work. The journey timeline starts with school and ends with the current student roles. The "short version" panel links to education, skills and leadership.

Education

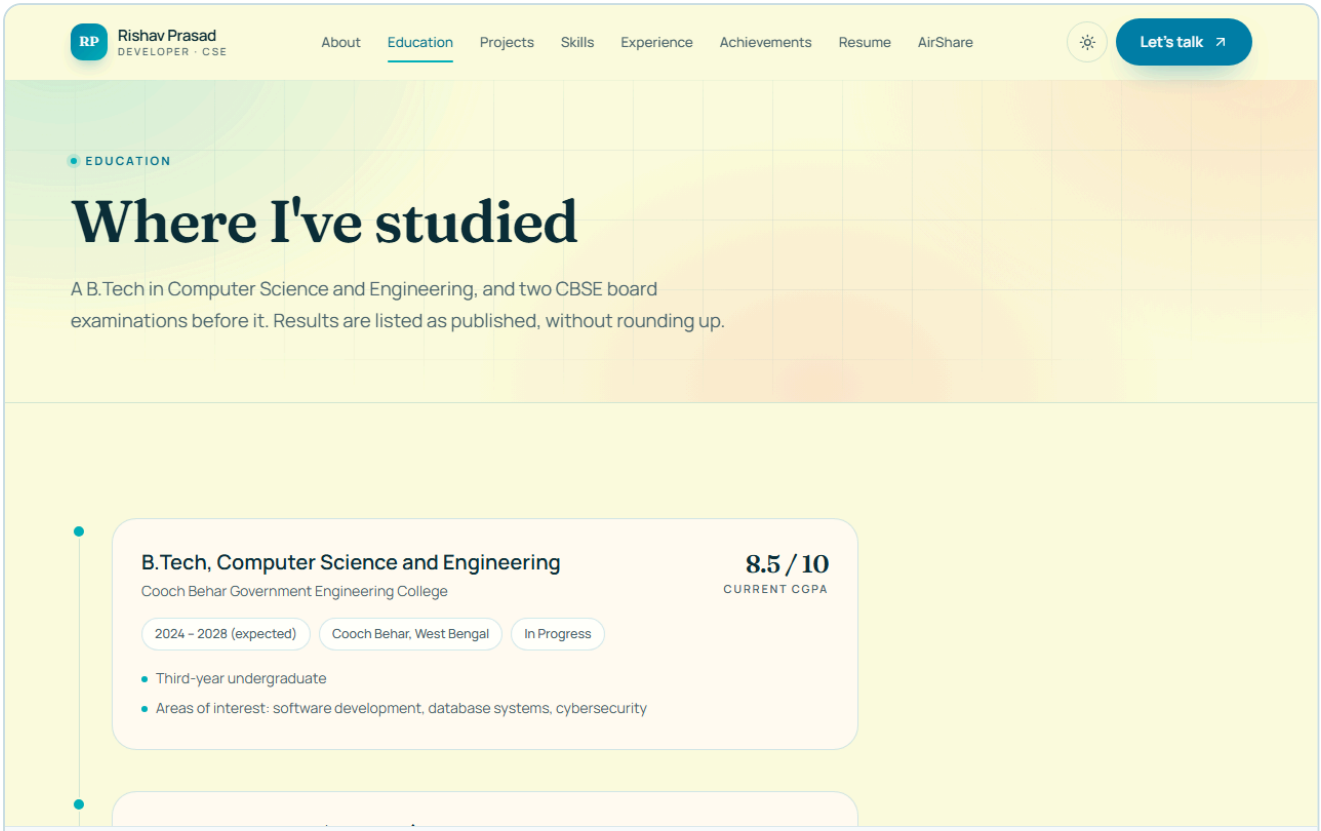


Figure 5. Education records: the current B.Tech and the two school certificates.

Three records are published: the B.Tech in Computer Science and Engineering at Cooch Behar Government Engineering College, Higher Secondary at Delhi Public School Siliguri, and Secondary at the same school. Each record shows the period, board or university where applicable, and the published score.

Skills

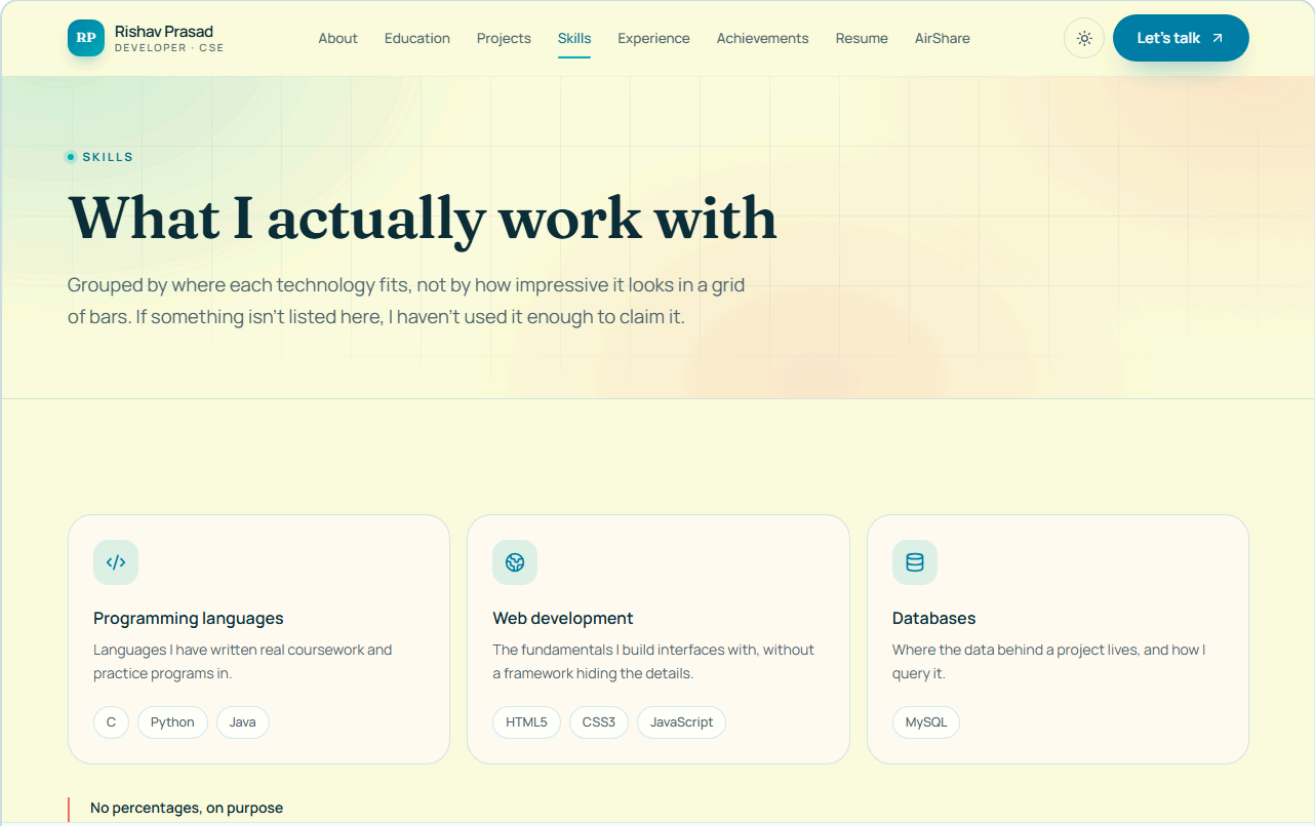


Figure 6. Skills are grouped into three cards, each with a short description and a list.

The skills page groups technologies by area rather than by proficiency percentage. It is a plain statement of the tools and topics the owner works with.

Projects

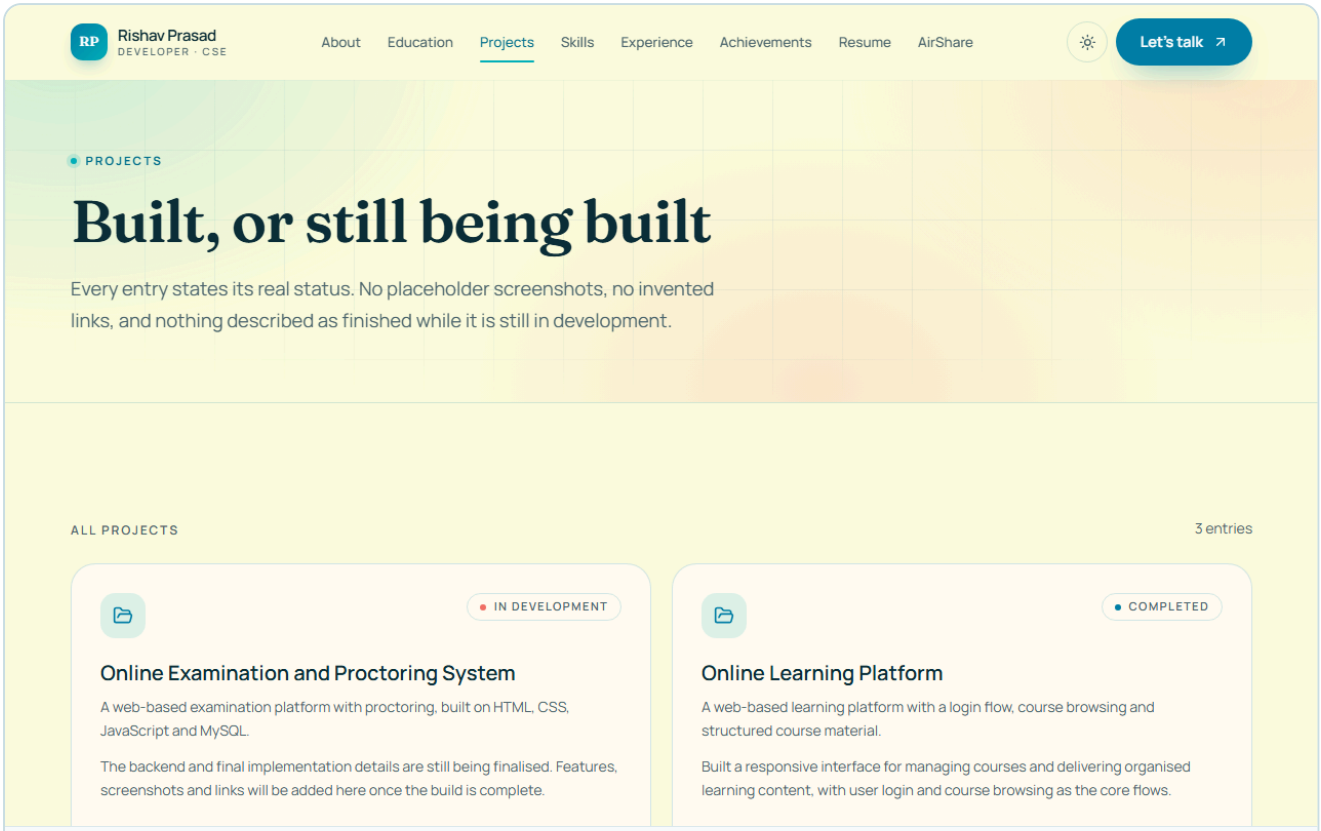


Figure 7. Three projects, each with status, summary, stack and optional links.

Each card shows a status (*In development*, *Completed* or *Shipped*), a summary, a longer detail paragraph, the stack and any links. Project screenshots are not part of the current content model; the cards are text-only by design.

Experience

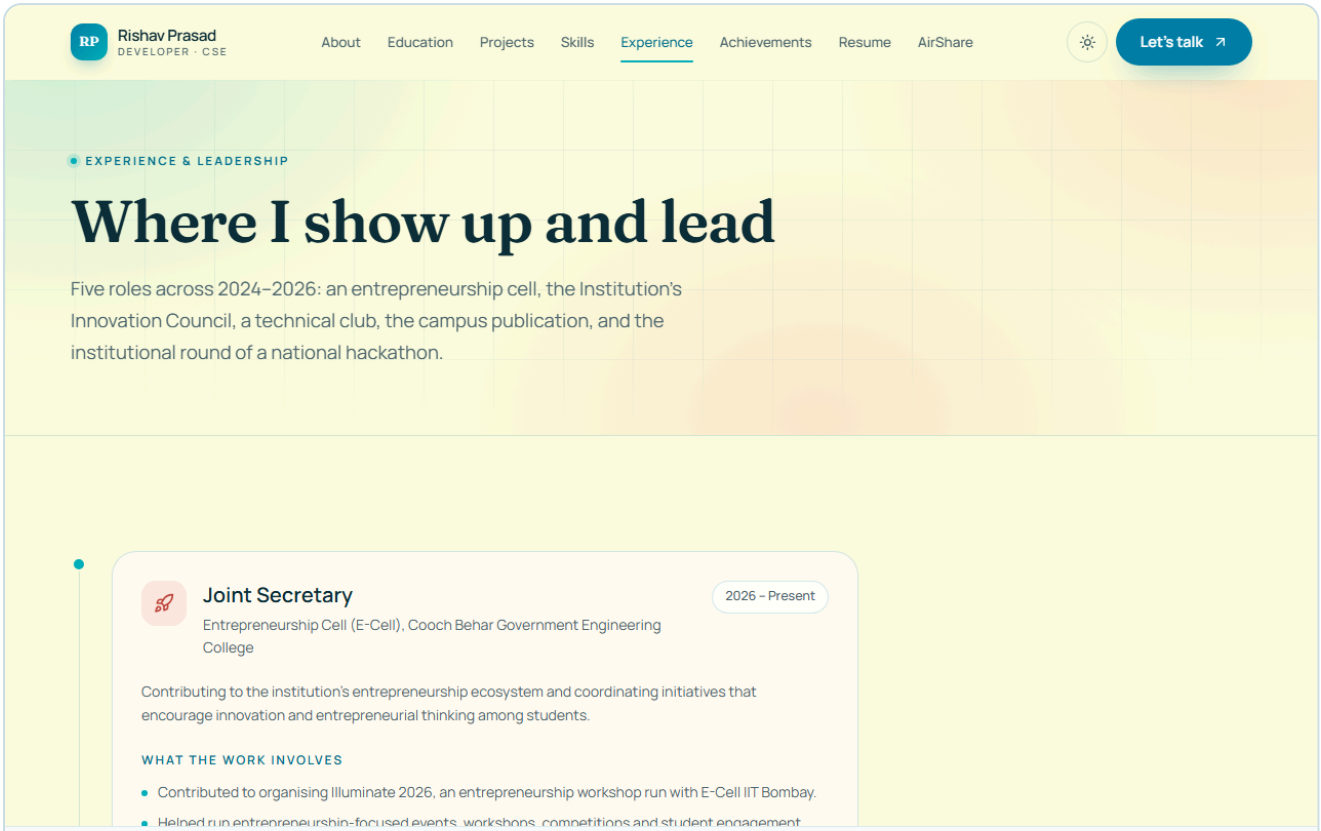


Figure 8. Leadership roles are presented as a timeline with responsibilities.

The experience page lists the five published leadership roles: Joint Secretary of the Entrepreneurship Cell, Joint Secretary of the Institution's Innovation Council, In-Charge of Tech-O-Nicks, Overall Coordinator of the institutional Smart India Hackathon 2026 round, and Reporter and Social Media Handler at CGEC Times.

Achievements

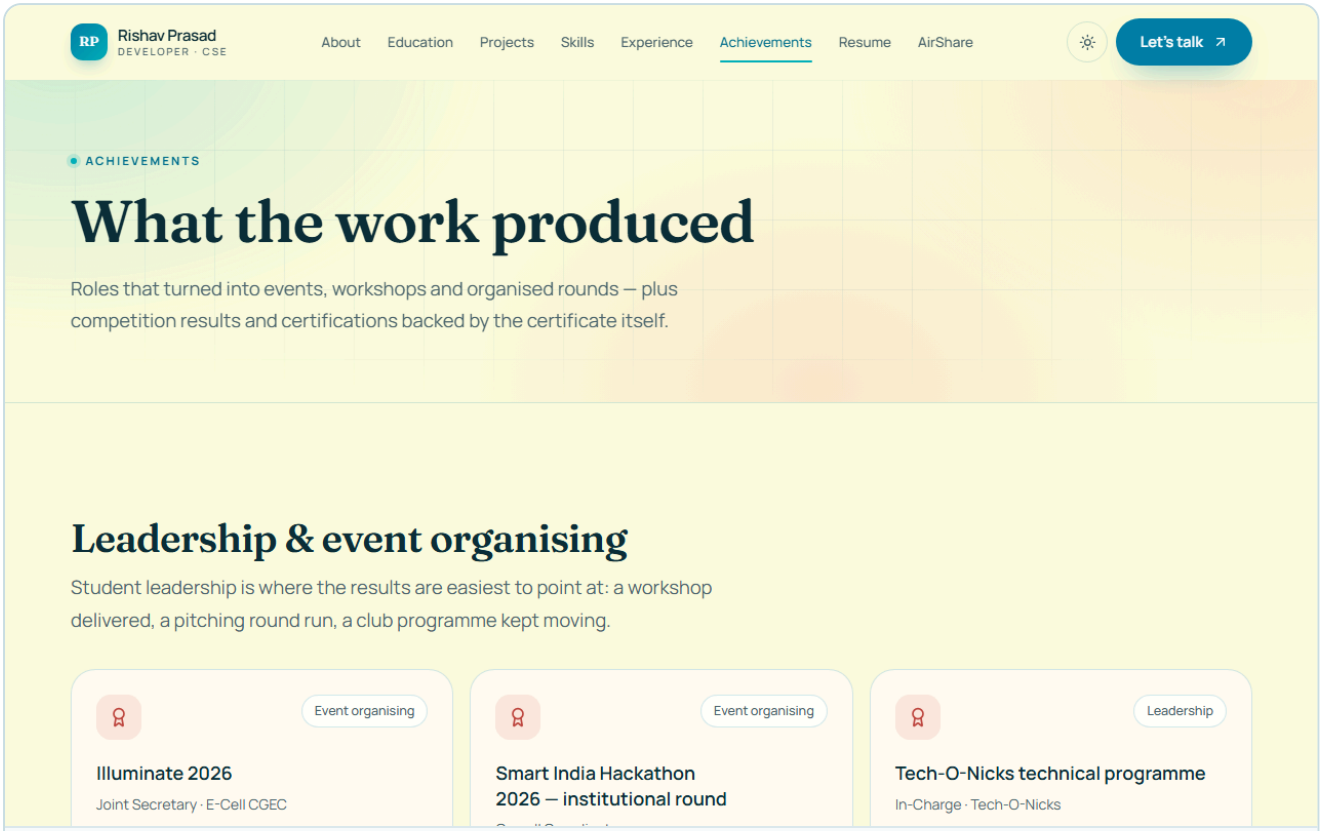


Figure 9. Leadership records, five awards and one certification with a certificate PDF.

This page separates leadership and event-organising records from awards. Awards include Esperanza 2025 (Robosoccer and Robosumo, winner), ByteBurst Hackathon 2025 (runner-up), Smart India Hackathon 2024 and 2025 (participant) and the Ideathon at Inzio, IIIT Guwahati (participant). The NPTEL certification “Programming, Data Structures and Algorithms using Python” links to a certificate PDF stored in the repository.

Resume

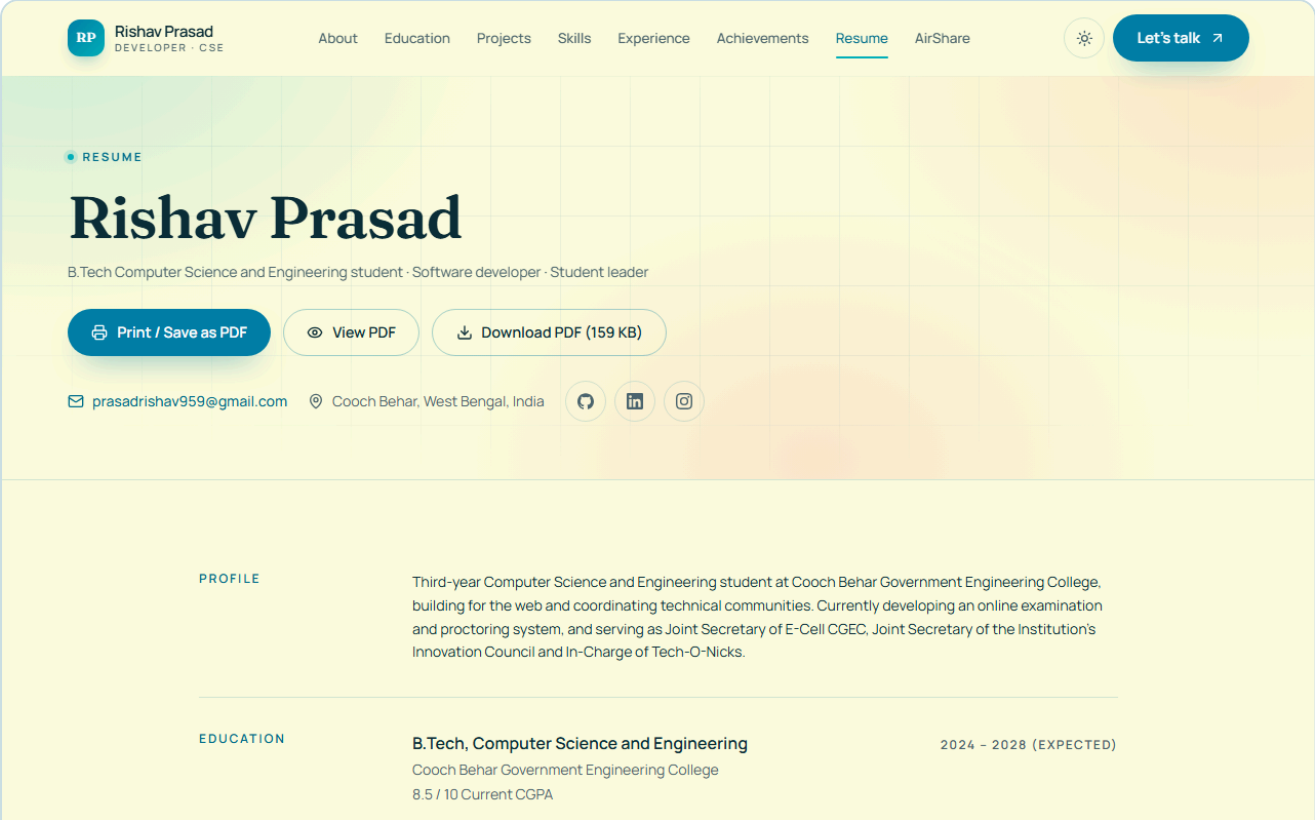


Figure 10. The resume page reproduces the approved details and offers the PDF.

The resume page shows the summary, education, experience, projects and certifications in a print-friendly layout. The “Download PDF” action serves `/resume/rishav-prasad-resume.pdf` from the static bundle. The page only exposes a PDF when the content flag `approvedForPublish` is true; it is currently true for the repository file.

Not found

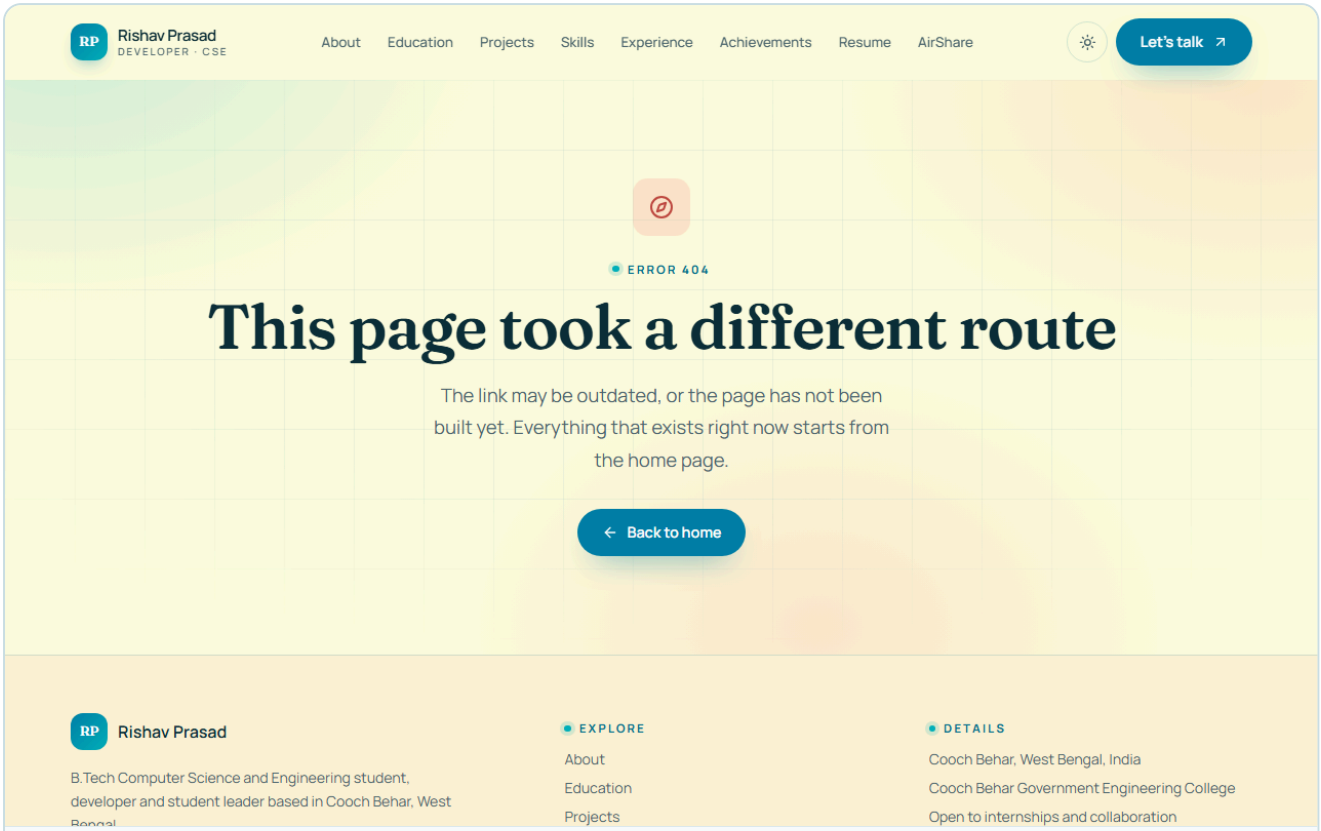


Figure 11. Unknown paths render the site's own 404 page, not a provider default.

The 404 page keeps the site header and footer and offers routes back to the main sections.

04

Contacting Rishav

The contact page offers two channels: a form stored in the site's database and a direct email link to **prasadrishav959@gmail.com**. No phone number is published.

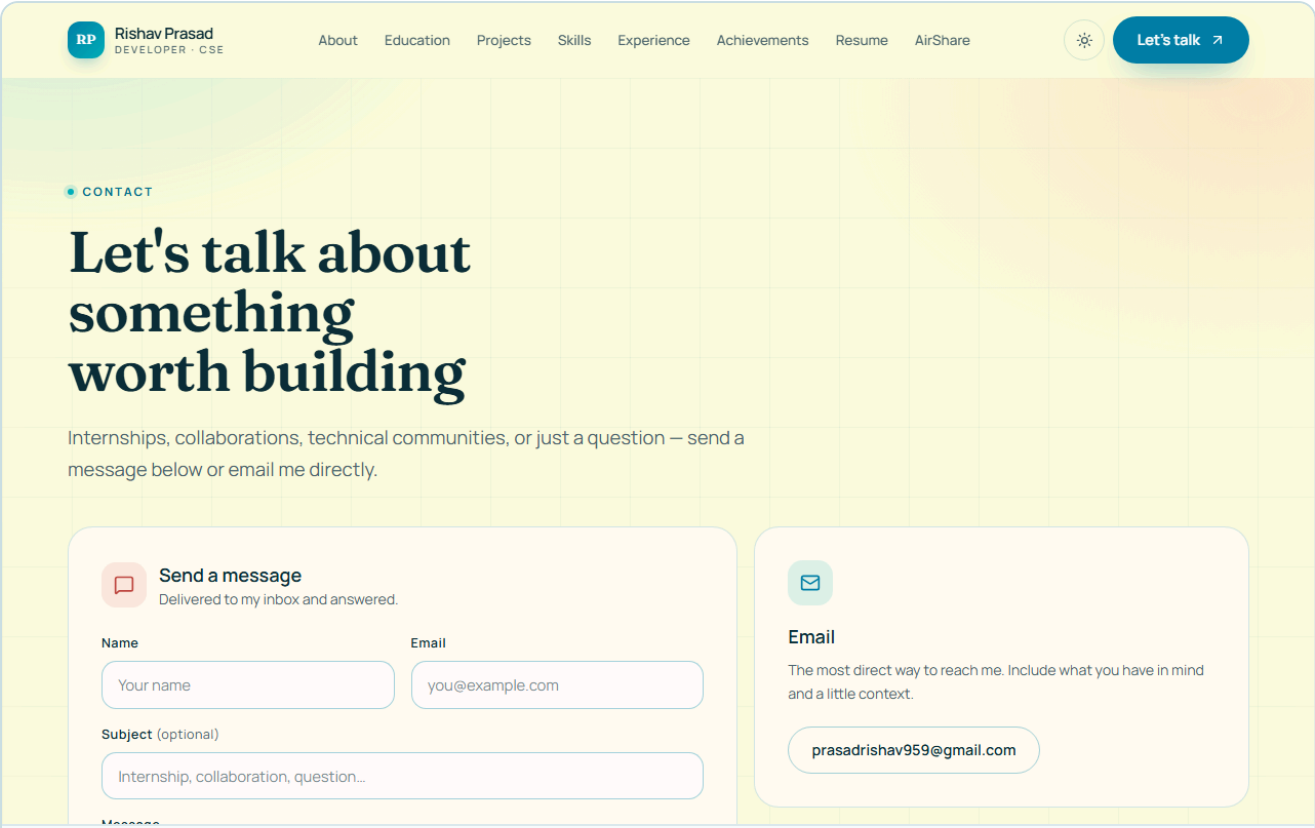


Figure 12. The contact form: name, email, subject, message and a Cloudflare Turnstile challenge.

What the form accepts

FIELD	RULE	REQUIRED
Name	2 to 80 characters	Yes
Email	Valid address, up to 160 characters	Yes
Subject	Up to 120 characters	No
Message	10 to 4000 characters	Yes
Verification	Cloudflare Turnstile token	Yes

What happens after sending

1. The browser sends the fields and the Turnstile token to `POST /api/contact`.
2. The Worker verifies the token with Cloudflare, rejects honeypot submissions silently, and validates every field.
3. A per-connection limit of five messages per hour is applied using a hash of the visitor's IP address; no raw IP is stored.
4. The message is stored in the D1 table `contact_messages` and appears in the owner's dashboard.
5. The visitor sees a confirmation. The owner sees the message the next time the Messages tab is opened.

A public submission was successfully stored on 31 October 2026, confirming the live path works. The sender's details are not reproduced in this documentation.

> **Note.** If the form says "The contact form is not configured yet", the Turnstile secret is missing on the Worker. Use the email link until the secret is set.

05

AirShare: local file transfer

AirShare sends files and short messages directly between two browsers on the same Wi-Fi. File bytes travel over WebRTC from browser to browser; the server only stores presence and connection metadata for a short time.

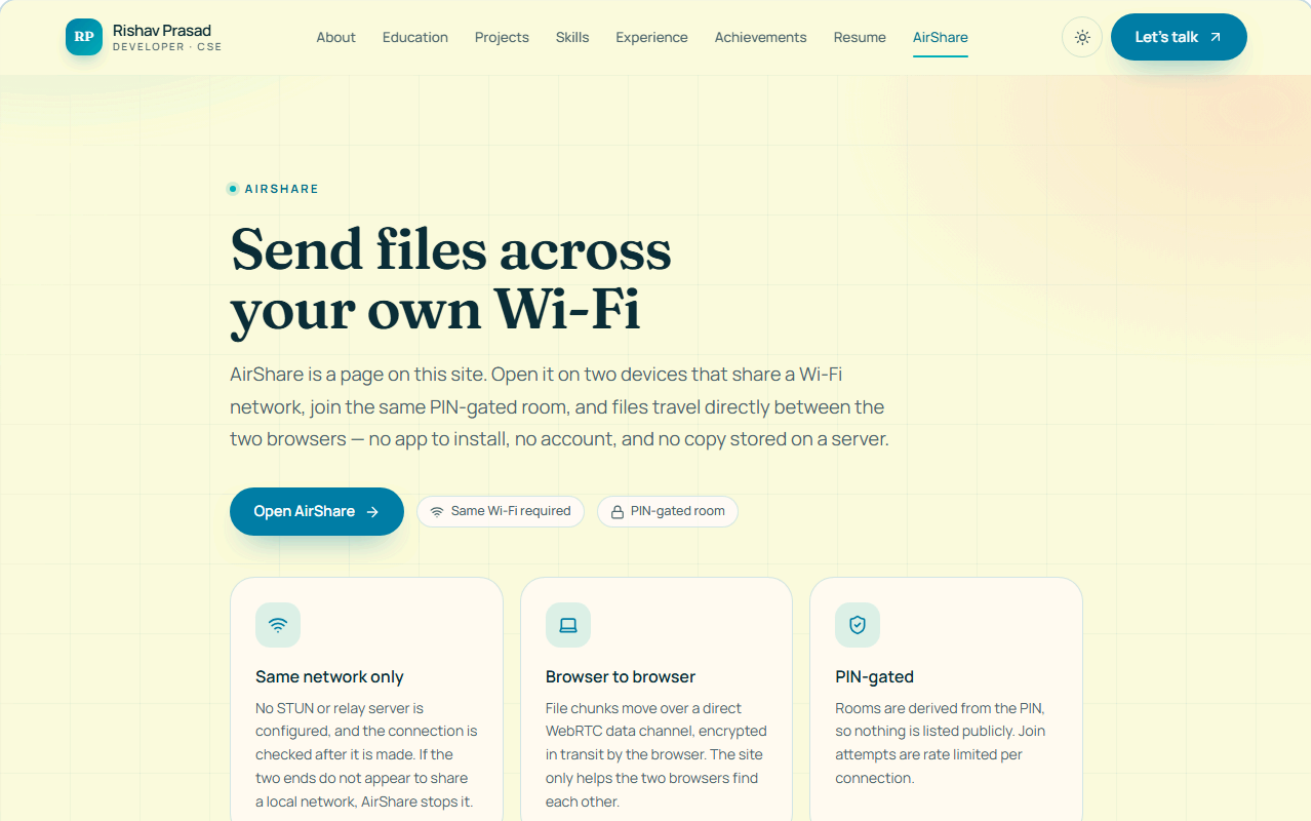


Figure 13. The guide explains the same-network requirement, the PIN model and the transfer steps.

Using AirShare

1. Connect both devices to the same Wi-Fi network. Guest networks that isolate clients will not work.
2. On the first device, open `rishavprasad.in/airshare/app/`, enter a device name and choose **Create a room**. AirShare generates a random 8-digit PIN.
3. On the second device, open the same page, enter a device name and the PIN, then choose **Join room**.
4. Wait until both devices show **Connected on this Wi-Fi**.
5. Choose a file (up to 100 MB) or type a message, then send it to the other device.
6. Keep both tabs open until the transfer completes. Use **Leave room** when finished.

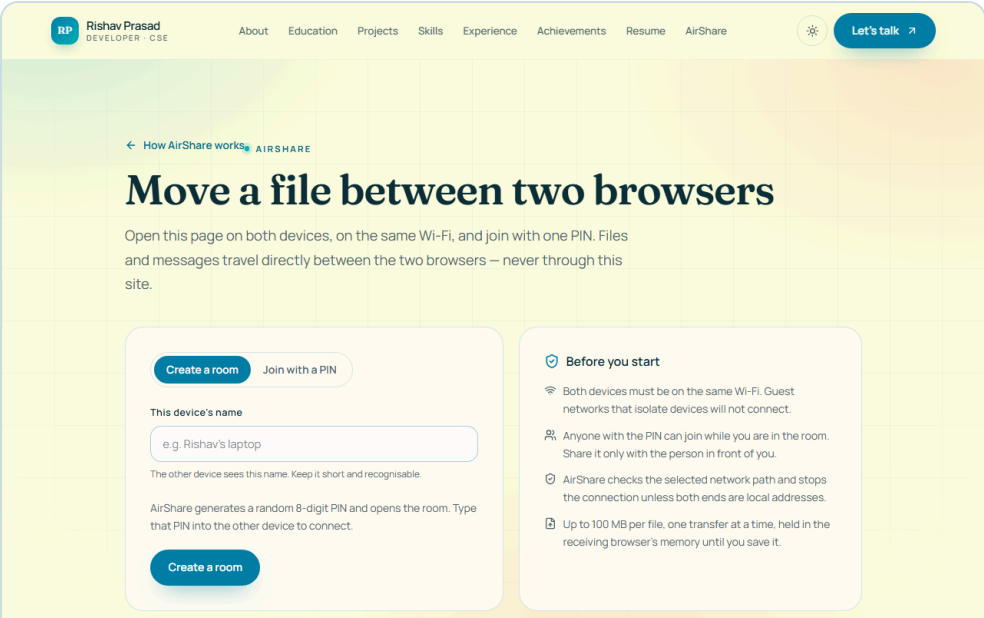


Figure 14. The AirShare app before joining: name, PIN entry and the two entry modes.

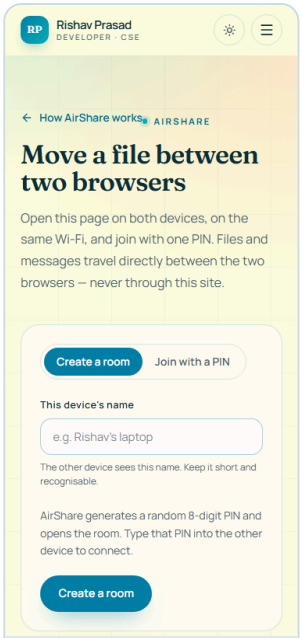


Figure 15. The same app at phone width.

Limits and behaviour

ITEM	LIMIT
PIN	6 to 10 digits; the app creates 8-digit PINs
Devices per room	8 simultaneous peers
File size	100 MB per file, sent in 64 KB chunks
Room lifetime	A peer disappears after 90 seconds without a heartbeat
Join limit	30 join attempts per 10 minutes per connection
Signalling	120 signals per minute per device; 64 KB per signal

AirShare deliberately has no STUN or TURN server and no relay. That is what keeps transfers local, but it also means some networks will refuse a direct connection. When that happens the app stops the connection and explains that a local connection could not be confirmed. This is a design limit, not a missing feature.

> **Warning.** Anyone who knows the PIN can join the room while it is open. Share the PIN only with the person receiving the file, and leave the room when the transfer is done.

06

The admin dashboard

The dashboard at `/admin/` is the owner's private editor for site content and the inbox for contact messages. It is excluded from the sitemap, marked `noindex`, and disallowed in `robots.txt`. Access still depends on the password, not on obscurity.

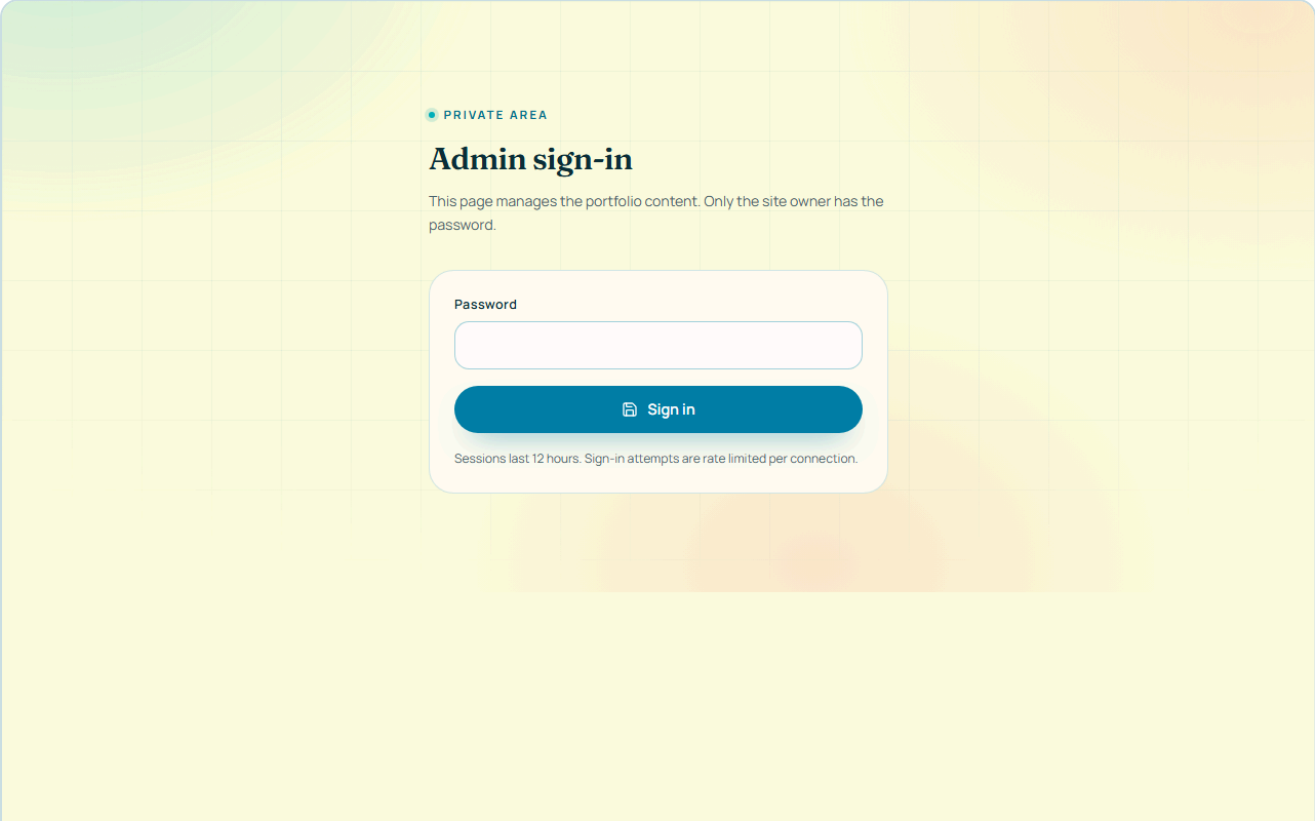


Figure 16. The sign-in screen. The signed-in dashboard is not reproduced here because it requires the owner's password.

Signing in

1. Open `rishavprasad.in/admin/`.
2. Enter the admin password and choose **Sign in**.
3. The Worker checks the password against a PBKDF2-SHA256 hash (100,000 iterations) and sets an HttpOnly, SameSite=Strict session cookie.
4. The session lasts 12 hours. Choose **Sign out** when finished, especially on a shared computer.

Wrong passwords are rate limited: eight failed attempts per 15 minutes per connection. A successful sign-in clears stale attempt records.

> **Important.** The admin password and the secrets behind it are not included in this package. The owner supplies `ADMIN_PASSWORD_HASH` and `GITHUB_TOKEN`; the hash is generated locally with `node scripts/hash-password.mjs` and installed with `wrangler secret put`. Never commit a password or paste it into a command that might fail mid-pipe.

What each tab edits

TAB	CONTENTS
Profile	Resume summary, focus domains, social profiles, factual details.
Education	Degrees, institutions, boards, scores and notes.
Skills	Skill groups with an icon, description and item list.
Experience	Leadership roles, periods, summaries and bullet points.
Projects	Title, slug, status, summary, detail, stack and links.
Achievements	Leadership records, awards and certifications.
Resume	Approval flag and the current PDF link; replacement upload is hidden while R2 is disabled.
Messages	The last 100 contact messages; mark as read or delete.
Media	Image library in R2; currently unavailable because no bucket is bound.

Editing and publishing

1. Edit fields in one or more tabs. Changes stay in the browser until saved.
2. Choose **Save changes** to write the whole document to D1. The API validates every field, allow-list and length limit before writing; a rejected document returns the list of problems and nothing is stored.
3. Choose **Publish** to ask GitHub to rebuild the static site. The Worker sends a `content-published` repository dispatch using the stored GitHub token.
4. Wait for the GitHub Actions run to finish. The public pages change only after that rebuild completes; this is usually a few minutes.

If the dashboard says “Publishing is not configured”, the GitHub token or repository variable is missing. If it says the rebuild request was rejected, the token has expired or lost permission to dispatch workflows.

> **Why two steps?** D1 is the source of truth for edits; the website is a static build. Saving alone does not change the public pages, and publishing without saving rebuilds the previous content. Always save first, then publish.

Messages and media

The Messages tab lists stored contact messages newest first, with an unread flag. Opening a message marks it read; deleting removes the row permanently. The Media tab and the resume upload require an R2 bucket binding that is currently commented out in `wrangler.jsonc`, so both show a configuration notice instead of an upload control. The resume PDF that the site serves lives in the repository at `public/resume/rishav-prasad-resume.pdf`.

Session and error states

MESSAGE	MEANING	ACTION
Admin login is not configured yet	`ADMIN_PASSWORD_HASH` or `SESSION_SECRET` is missing.	Set both as Worker secrets.
Invalid credentials	Wrong password.	Retry; after eight failures wait 15 minutes.
Too many attempts	Login rate limit reached.	Wait, then sign in.
Your session expired	The 12-hour cookie expired or the secret changed.	Sign in again; unsaved edits are lost.
Stored content could not be read	D1 returned invalid content; the editor refuses to overwrite it.	Restore from backup before saving (see document 05).

07

Appearance and accessibility

The site ships a light and a dark theme built from the same “Summer Dream” palette. The theme follows the operating system on first visit, then remembers the visitor’s choice in `localStorage` under the key `theme`. A toggle in the header switches modes without a page reload.

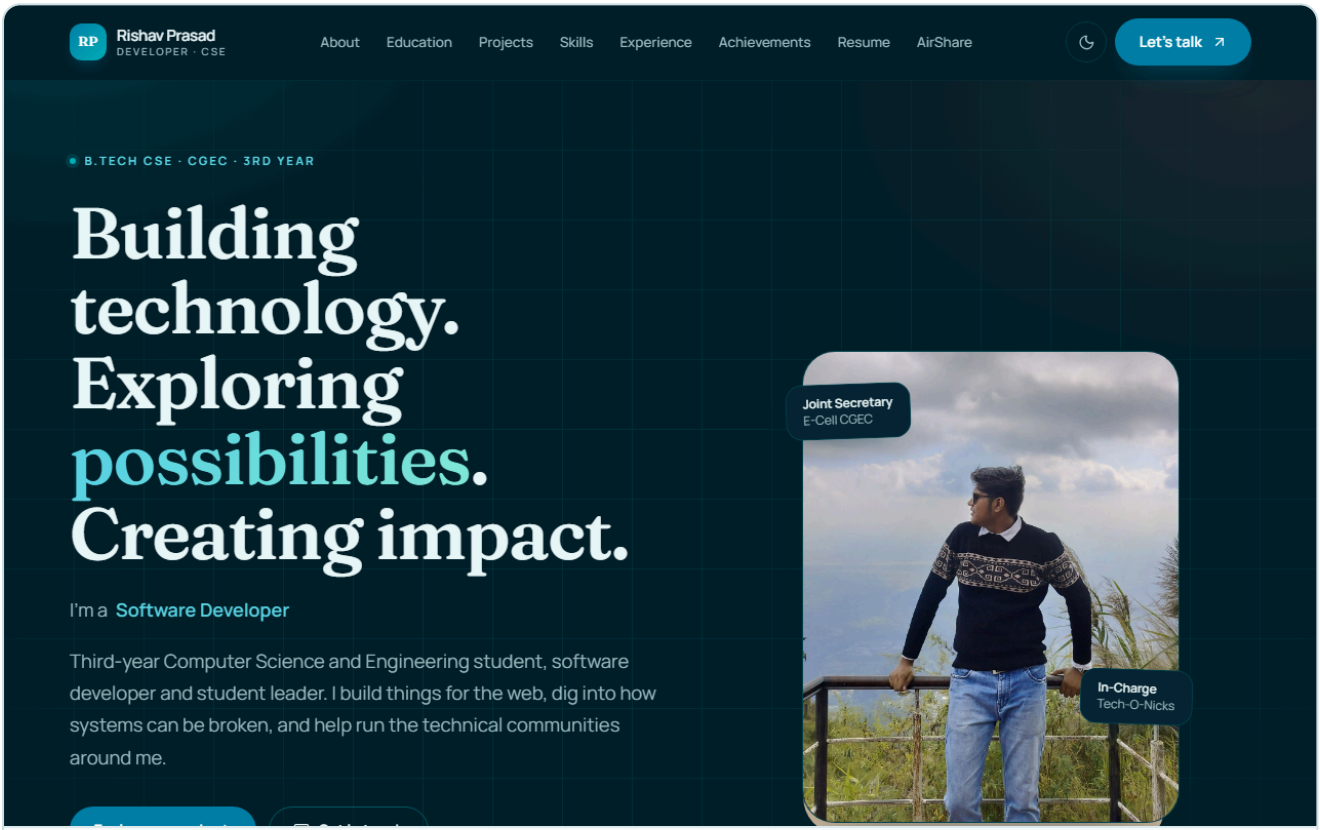


Figure 17. The dark theme uses the same accents on a deep teal background.

Accessibility measures

- A “Skip to content” link is the first focusable element.
- The navigation toggle exposes its state, and the mobile menu is keyboard operable.
- Every form control has a visible label; the contact form uses native validation hints.
- Images carry alternative text, and decorative elements are hidden from assistive technology.
- Focus outlines are preserved and contrast is designed for both themes.
- The admin editor labels its inputs with real `label` elements, including dynamically generated lists.

Tested widths and browsers

Automated checks cover 320, 375, 425, 768, 1024, 1440 and 1920 px across all eleven public routes. Mobile Safari on a physical phone has not been re-run in this engagement; the layout was verified in Chromium-based automation. AirShare needs WebRTC, which current Chrome, Edge, Firefox and Safari provide.

08

Troubleshooting

SYMPTOM	LIKELY CAUSE	WHAT TO DO
Contact form says verification failed	The Turnstile challenge expired or the widget did not load.	Reload the page, complete the challenge again, or use the email link.
“Too many messages from this connection”	Five submissions in one hour from the same network.	Wait, or send email directly.
AirShare never shows “Connected”	Guest Wi-Fi isolation, different networks, or a network that blocks direct peer connections.	Use the same non-isolated Wi-Fi; retry; if it still fails, use a cable or a local transfer tool.
“AirShare stopped this connection because the two devices did not appear to share a local network”	WebRTC selected a public or relay-like path, which AirShare rejects on purpose.	Reconnect both devices to the same router and retry.
Admin sign-in says “Something went wrong”	An unexpected server error, often a missing secret or a database problem.	Check `/api/health`; if the database is unavailable, consult document 05.
Saved changes do not appear on the site	Publish was not run, or the rebuild is still queued.	Open the GitHub Actions run, confirm success, then reload with a hard refresh.
Resume download returns 404	The approved file path is wrong or the build has not run since the file changed.	Restore `public/resume/rishav-prasad-resume.pdf` and redeploy.

09

Privacy and data

The public site sets no advertising or tracking cookies. Cloudflare Web Analytics is enabled at the zone level; it is cookieless and aggregates page views. The only first-party storage in the browser is the theme preference. AirShare keeps a device name in `localStorage` so the field is prefilled.

Contact messages contain the name, email, optional subject and message text supplied by the sender, plus a salted hash of the connection IP for rate limiting. Messages are visible only in the authenticated dashboard. AirShare rooms store a device name, a peer identifier, a hashed room identifier and short-lived WebRTC signalling payloads; file contents never reach the server. Peer and signal rows are pruned automatically.

10

Glossary

D1	Cloudflare's SQLite-based database used for content, messages and AirShare signalling.
Worker	The Cloudflare serverless function that serves `/api/*` and `/media/*` and falls back to static assets.
Static export	The pre-rendered HTML, CSS and JavaScript written to `out/` by `next build`.
Turnstile	Cloudflare's CAPTCHA alternative used to protect the contact form.
PBKDF2	The password hashing function used for admin sign-in; Workers cap it at 100,000 iterations.
Repository dispatch	A GitHub API event that starts the rebuild workflow after a publish action.
R2	Cloudflare object storage. It is optional here and currently not enabled.
WebRTC	The browser API that carries AirShare files directly between devices.